Bellman Equation

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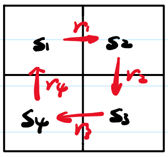
Return: Discounted sum of rewards along a trajectory.

Discount rate

policy prob?



$$\begin{cases} v_1 = r_1 + \gamma r_2 + \gamma^2 r_3 + \gamma^3 r_4 + \cdots = r_1 + \gamma(r_2 + \gamma r_3 + \cdots) = r_1 + \gamma v_2 \\ v_2 = r_2 + \gamma v_3 \\ v_3 = r_3 + \gamma v_4 \\ v_4 = r_4 + \gamma v_1 \end{cases}$$

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{pmatrix} + \gamma \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix}$$

$$\Rightarrow \quad v = r + \gamma P v \quad \Rightarrow \quad v = (I - \gamma P)^{-1} r$$

a very simple bellman-equation.

State Value:

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1}$$

random variables!

Policy: $\pi(\cdot|s)$

① $S_t \longrightarrow A_t$: $\pi(A_t|S_t) = P(A_t = a | S_t = s)$

② $S_t, A_t \to R_{t+1}$: $p(R_{t+1} = r | S_t = s, A_t = a)$

③ $S_t, A_t \to S_{t+1}$: $p(S_{t+1} = s' | S_t = s, A_t = a)$

Distributions!

Trajectory: $S_t \xrightarrow{A_t} R_{t+1}, S_{t+1} \xrightarrow{A_{t+1}} R_{t+2}, S_{t+2} \xrightarrow{A_{t+2}} \cdots$

Return: $G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots$

Def (State value):

$$\boxed{v_\pi(s) = E[G_t | S_t = s]}$$

$v(\pi, s)$

Expectation of possible returns

Bellman Equation:

Since

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = R_{t+1} + \gamma G_{t+1}$$

therefore

$$v_\pi(s) = E[G_t | S_t = s] = E[R_{t+1} + \gamma G_{t+1} | S_t = s] = E[R_{t+1} | S_t = s] + \gamma E[G_{t+1} | S_t = s].$$

calculate separately

$$E[R_{t+1} | S_t = s] = \sum_a \pi(a|s) \sum_r p(R_{t+1} = r | S_t = s, A_t = a) r$$

$$E[G_{t+1} | S_t = s] = \sum_{s'} E[G_{t+1} | S_t = s, S_{t+1} = s'] p(S_{t+1} = s' | S_t = s)$$

$$E[G_{t+1} | S_t = s] = \sum_{s'} E[G_{t+1} | S_t = s, S_{t+1} = s'] p(S_{t+1} = s' | S_t = s)$$

$$\downarrow \text{MDP}$$

$$= \sum_{s'} E[G_{t+1} | S_{t+1} = s'] p(S_{t+1} = s' | S_t = s)$$

$$= \sum_{s'} v_\pi(s') \sum_a \pi(a|s) p(S_{t+1} = s' | S_t = s, A_t = a)$$

$$\Rightarrow v_\pi(s) = \sum_a \pi(a|s) \sum_r p(R_{t+1} = r | S_t = s, A_t = a) r + \sum_{s'} v_\pi(s') \sum_a \pi(a|s) p(S_{t+1} = s' | S_t = s, A_t = a)$$

Thy. (Bellman Equation)

$$\boxed{v_\pi(s) = \sum_a \pi(a|s) \Big(\underbrace{\sum_r p(r|s,a) r}_{\text{immediate reward}} + \underbrace{\gamma \sum_{s'} p(s'|s,a) v_\pi(s')}_{\text{future return}} \Big), \quad \forall s \in S.}$$

① Calculate $v_\pi(s)$ & $v_\pi(s')$: Bootstrapping

② $\pi(a|s)$: given policy evaluation

③ $p(r|s,a)$, $p(s'|s,a)$: dynamic model.

Matrix-vector form of the Bellman Equation

For state set :

$$S = \{s_i\}_{i=1}^n .$$

Rewrite Bellman:

$$v_\pi(s_i) = r_\pi(s_i) + \gamma \sum_{s_j} p(s_j|s_i) v_\pi(s_j)$$

$$\Rightarrow \boxed{v_\pi = \begin{pmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ \vdots \\ v_\pi(s_n) \end{pmatrix} = r_\pi + \gamma P_\pi v_\pi}$$

$$\underbrace{[P_\pi]_{ij}}_{\text{State transition matrix}} = p_\pi(s_j|s_i)$$

Solve State-value (Policy eval)

① Closed-form :

$$v_\pi = (I - \gamma P_\pi)^{-1} r_\pi .$$

practical

② Iterative solution :

$$v_{k+1} = r_\pi + \gamma P_\pi v_k$$

init $v_0 \Rightarrow v_1 \Rightarrow v_2 \Rightarrow \cdots \Rightarrow v_n$

$n \to \infty \Rightarrow v_n \to v_\pi$.

* Action value

state + action.

Def (Action value)

$$\boxed{q_\pi(s,a) = E[G_t | S_t = s, A_t = a]}$$

from :

$$v_\pi(s) = \sum_a \pi(a|s) q_\pi(s,a)$$

derive :

$$\boxed{q_\pi(s,a) = \sum_r p(r|s,a) r + \gamma \sum_{s'} p(s'|s,a) v_\pi(s')}$$

derive :

$$\boxed{q_\pi(s,a) = \sum_r p(r|s,a)\, r + \gamma \sum_{s'} p(s'|s,a)\, v_\pi(s')}$$